Yasser Nabil

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Education

Wuhan University *Wuhan, China*

B.Sc. in Software Engineering *Sep 2018 - Jun 2022*

- **Grade:** 3.95/4.00 GPA
- Focused in software skills including **Python, C++, C, Java, and HTML/CSS**, with a strong understanding of **databases** and **data structures**, as well as efficient programming techniques.

Università Degli Studi Di Padova *Padua, Italy*

M.Sc. in Computer Science *Sep 2022 - Jul 2024*

- **Grade:** 110/110
- Focused on applying theoretical knowledge to solve **real-world software engineering challenges**, enhancing my ability to **design** and **optimize** efficient, **scalable systems**.

Skills

Programming	Python, C/C++, Java, JavaScript, Typescript, Linux, SQL, Html, CSS
Explored skills	C#, Ms Azure, Jenkins, SASS, MATLAB
Libraries	Bootstrap, Node.js, React.js, Express.js, Selenium, pytest, FastAPI, Google Maps API
Technology	Git, AWS, Angular, Docker, Firebase, Kubernetes, Puppeteer, JupyterLab, OracleDB, MongoDB, MySQL, Redis
Languages	French, English, Arabic, Russian, Italian, Chinese

Work Experience

Evertz *Burlington, ON, Canada*

Intermediate Software Engineer *May 2026 - Present*

Software Engineer *Nov 2024 - May 2026*

- Automated internal tasks using **Python** (cron, subprocess), saving ∼8 hrs/week per team and streamlining cross-team workflows.
- Built internal tools with **React.js** (Hooks, Context API) to help team members easily interact with internal systems and resolve issues without technical blockers.
- Set up and maintained **Docker**-based dev/staging environments and **Redis** caching queues; reduced onboarding time from hours to 15 minutes.
- Wrote automated tests with **Pytest** and **Selenium**, increasing test coverage by 60% and reducing post-release bugs significantly.
- Used **Linux** daily for scripting (Bash, udev, systemd).

The European Organization for Nuclear Research (CERN) *Geneva, Switzerland*

FULL STACK DEVELOPER *Jun 2022 - Sep 2022*

- Migrated legacy notebook infrastructure to a modern, modular setup using **JupyterLab**, simplifying future feature integration.
- Developed and maintained both front-end and back-end components within **JupyterLab**, enabling smooth delivery of 5+ new internal tools.
- Implemented a custom download mechanism in **Python** and built interactive frontend components with **TypeScript**, **HTML/CSS**, and **JavaScript** for the **MkDocs** interface, reducing user confusion and support requests by over 40%.
- Created automated test units using **Puppeteer**, then deployed the stable release using **Docker** and **Kubernetes**, reducing deployment time by 60% and ensuring zero-downtime updates.

Orange - Cloud Whale *Casablanca, Morocco*

SOFTWARE ENGINEER *Dec 2021 - Jun 2022*

- Managed over **300,000** records using **OracleDB**, optimizing queries improving data retrieval performance and system responsiveness.
- Automated routine system tasks with custom **Linux** scripts, reducing manual intervention and saving several hours of operations time.
- Developed client-focused **Python** scripts to help account management workflows.

DIGI4 *Casablanca, Morocco*

FULL STACK DEVELOPER *Sep 2020 - Mar 2021*

- Built full-stack features using **Angular** and **Node.js**, contributing to both client-facing UI and backend services.
- Developed responsive UI components using **Angular** (TS, HTML, CSS), enhancing user engagement and overall interface usability.
- Led the design and implementation of a comment and ranking system for the client's website, following an **Agile** development process.
- Engineered an automated location system using **Node.js** and the **Google Maps API** to accurately locate over **100,000** stores.
- Collaborated closely with clients to gather feedback and propose optimal solutions, then deployed services using **AWS** infrastructure.
- Wrote a data enrichment script to automatically insert geo-coordinates and other metadata into the **MongoDB** database, improving data quality and searchability.

Projects

WebChat App *Angular, Node.js, Vue.js*

- I developed a local web chat application enabling users to engage in virtual conversations.
- Using **Node.js** for the backend and Angular for the frontend, I assured perfect functionality.
- Users can engage in real-time messaging similar to conventional messaging platforms, using a local internet connection.
- The project includes a comprehensive documentation available on my **GitHub page**.

Meta Language Interpreter *F#, Functional Languages*

- The aim of this project is to implement the algorithm of type inference.
- Implemented with a type inference mechanism to ensure absolute type safety within expressions.
- Developed comprehensive understanding of the functioning of programming language compilers.
- **GitHub repo:** TinyML.

SWAN Gallerie - Jupyter Extension *Python, Typescript, Docker, Kubernetes*

- Migrate the main structure of SWAN CERN using a more stable framework.
- Created new extensions to the webapp using **Typescript**, **Docker**, **Kubernetes**.
- Worked from development to deployment of the app within CERN's network.
- **GitHub repo:** Yasser-SwanGallery-Extension.

Thesis

An Empirical Study of Decentralized Fine-tuning for Machine Learning Models *Python, Pytorch, Scikit-Learn*

MSc Thesis - https://hdl.handle.net/20.500.12608/68876

- Made an empirical study to determine the best models to use in different situations.
- Extensive use of ML lybraries **(Pytorch, Numpy, Scikit-learn, matplotlib)**
- Using **Python Notebook** I simulated Gossip Learning and Federated Learning.
- I experimented with **ResNet18** and **ShuffleNetv2** under various conditions to mimic real-world situations.
- The project includes a comprehensive documentation available on the **Unipd Database**.

Publication

Integration of Jupyter Notebook Gallery with JupyterLab Interface of SWAN - Service for Web Based Analysis *Sep 2022*

CERN SWAN, Service for Web based ANalysis, is a platform to perform interactive data analysis in the cloud. It helps the user to analyze data straightforwardly without the need to install any software. The current SWAN utilizes a structure based on Jupyter Notebooks that is working accurately. However, by migrating the old infrastructure to JupyterLab the user will benefit from the maneuverability of the service as well as the possibility for many new features and chances for improvement. The new structure, thanks to its modular structure allows for the possibility to browse through the different files the user possesses while being able to view the different notebooks within the same window. SWAN Gallery, on another hand, is a set of examples for CERN SWAN to help users see working examples and be able to practice with them. The project's goal consists of implementing this static website as an extension in the new infrastructure using JupyterLab. The process containing the different steps along any problem encountered during the finalization of the project will be correctly explained and stated throughout the report.